

Thermodynamic response of gas injection-and-withdrawal process in salt cavern for underground gas storage

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HIGHLIGHTS

- Analytical solution to temperature and pressure variation in gas injection-and-withdrawal process was derived.
- The mathematical model is employed to calculate the thermal stress.
- Fully coupled thermal-mechanical numerical simulations are performed.
- Fast gas-withdrawal rate induces more tensile stress concentrations.

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ABSTRACT

Cycled mode of operation leads to temperature and pressure variation in salt cavern for underground gas storage due to mechanical and thermal loading. In this study, based on metamorphic thermodynamic principle, a mathematical model of thermal analysis is proposed for gas injection-and-withdrawal process, formulates a newly analytical solution to temperature and pressure variation with time during injection-and-withdrawal process. The proposed solution is validated with the withdrawal test results. The derived solution can be set as boundary conditions for the thermal-mechanical numerical modeling, which is capable of calculating thermodynamic response for both constant and in-constant air mass flow rate. Fully coupled thermo-mechanical numerical simulations are performed to evaluate thermal effects on cavern wall at different gas withdrawal and injection rates. The results indicate that during process of gas-withdrawal, fast withdrawal rate leads to increase of tensile stresses, and there are distinctive tensile stress areas on the roof and floor of the cavern; while in the gas-injection period, there are no tensile stress area, however, it inclines to occur stress disturbance at intervention of interlayers. Parameters analysis shows that both the thermal coefficients and variation of injection-and-withdrawal rates have impacts on the thermodynamic responses of surrounding rock mass.

1. Introduction

Salt caverns, serves as storage vessels, in functional performance of high rates of injection and withdrawal [6]. Nowadays, the surge of natural gas consumption in China demands greater on peak-shaving and supply capacity of Underground Gas Storage (UGS) [18]. Since the salt cavern gas storage has the advantages of economic safety, less required cushion gas and providing very high withdrawal and injection rates, the research on the salt cavern gas storage is gradually deepening.

Salt rock, as evaporates, are the weakest rock, with the favorable creep behavior, subjected to temperature influence [7,11,19]. Thermal tensile

stresses may induce fractures at the wall of salt cavern by thermos-mechanical loading [2,15,17]. The experiment investigation into rapid cooling of rock salt indicates that rapid depressurization result in tensile fracturing [33]. The fracture damage development from progressive microcracking could ultimately lead to spalling at salt-rock of cavern wall [5,9,13]. Therefore, the potential hazard resulting from thermal fractures must be concerned for design criteria, and operation safety [8,10].

In underground gas storage, the temperature and pressure distribution inside the salt cavern is a complicated problem. The influences of temperature variation over time play an important role in underground gas storage in salt cavern. The consequences of thermodynamic

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Nomenclature

A	cavern wall surface area, m ²
c_p	constant pressure specific heat, J/(kg•K)
c_v	constant volume specific heat, J/(kg•K)
h	specific enthalpy of air, J/kg
h_0	heat transfer coefficient, W/(m ² K)
K	coefficient of temperature difference between air and cavern wall
M	air mass flow rate, kg/s
m	air mass flow rate, kg/s
Q	convective heat transfer rate, W
R	Specific air constant, J/(kg K)
T_w	cavern wall surface temperature, K
U	Internal energy, W
V	cavern volume, m ³

Greek symbols

γ_1	withdrawal rate, m ³ /s
γ_2	injection rate, m ³ /s
τ	process of injection or withdrawal duration time, s
$\frac{v_r^2}{2}$	Kinetic energy in unit time flow in-out cavern, J/kg
ρ	air density, kg/m ³
k_e	Slope of withdrawal rate
c_e	Initial withdrawal rate, m ³ /s

Subscripts

0	Initial state
R	Rock
in	inlet
e	outlet

processes are: the increasing and decreasing of gas temperature during injection and withdrawal; heat exchanges between the gas and surrounding rock salt; and the thermal conduction within the salt formation mass. The consequences of thermal effects are more obvious in UGS, with depth over 2000 m due to increasing geothermal gradient. The convergence considering thermal effects is analyzed quantitatively [14]. The rapid injection and withdrawal cycles are always accompanied by rapid fluctuations in the internal pressure of salt cavern, causing temperature variation susceptible to damage the rock salt formation [1]. In process of gas injection-and-withdrawal in salt cavern, the temperature and pressure variations are significant factors that affect the overall operating performance of UGS. Xia et al. analyzed the thermodynamic performance of CAES system, and proposed a diabatic analytical solution in simplified and unified solution, considering the heat exchange between cavern air and the surrounding environment. Four parameters mainly caused the errors of the proposed solution are investigated in this study, respectively which are: the rock thermal diffusivity, the heat transfer coefficient, the difference between injected air temperature and the initial air temperature, and the injected to initial cavern air mass ratio.

The operation safety of UGS as one of the key perspectives has aroused increasing concerns. However, most of the stability evaluation of salt cavern is based on the static behavior of salt cavern under static loading condition, and those pertaining parameters, such as, geologic parameters, geometric parameters, and rock engineering parameters [12]. Although the temperature effects are analyzed, there is limited study on stability of UGS, particularly considering the thermal stress induced by thermal effects during gas injection-and-withdrawal process on surrounding rock salt formation. This paper presents a new analytical solution to air variations of temperature and pressure in the salt cavern, and thermal-mechanical modeling for thermodynamic response of underground salt cavern storage during operation process of gas injection-and-withdrawal. This fully coupled thermal-mechanical numerical model is used to simulate the thermos-response of salt cavern wall during injection-and-withdrawal process.

2. Analytical modeling of Injection-and-Withdrawal process

2.1. Governing equations

During the gas injection-and-withdrawal operation in salt cavern,

air flows in-and-out of the cavern system, the mass and energy exchange at the cavern port. The thermodynamic process is governed by the following equations:

Gas state equation:

$$pV = mRT \quad (1)$$

Mass conservation equation in representation of the instantaneous rate of change:

$$\frac{\delta m_i}{d\tau} - \frac{\delta m_e}{d\tau} = \frac{dm}{d\tau} \quad (2)$$

Energy conservation equation:

$$\frac{\delta m_i}{d\tau} - \frac{\delta m_e}{d\tau} = \frac{dm}{d\tau} \quad (3)$$

$$\frac{\delta Q}{d\tau} + \sum \left(h + \frac{v_r^2}{2} \right) \frac{\delta m}{d\tau} = \frac{dU}{d\tau} \quad (4)$$

Heat exchange between gas in cavern and surrounding rock in a time period of $d\tau$:

$$\delta Q = h_c A (T_w - T) d\tau$$

局限性：理想气体
纯净气体

In this study, we derive the analytical solution for temperature and pressure variation during the dynamic process of gas injection-and-withdrawal in underground salt cavern. To obtain a simplified and unified analytical solution, some fundamental assumptions for gas injection-and-withdrawal in salt cavern are summarized as follows:

1. the salt cavity is filled with dry, clean natural gas in uniform distribution, i.e. ideal gas
2. in the process of gas injection-and-withdrawal, the compositions of gas keep constant at each moment.
3. the flow in-and-out is steady flow process, the loss at wellbore and surface facilities can be neglect.
4. The salt-cavern is not newly-implemented. The rock mass cooling at the salt cavern induced by cavern leaching has not been taken into account.

Fig. 1 shows the simplified model of salt cavern for gas storage, the shape of cavern is assumed as spherical. In this study, the salt cavern and the surrounding rocks is an integrated system. The salt cavern is considered a control volume system. For this open system, there is heat

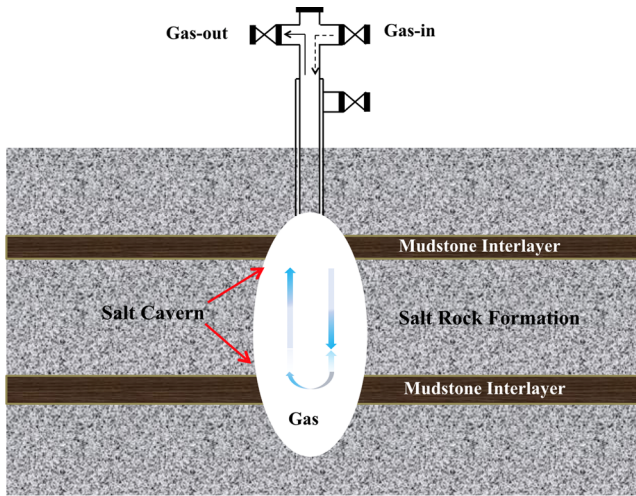


Fig. 1. Schematic diagram of underground gas storage in salt cavern with interlayers of mudstone.

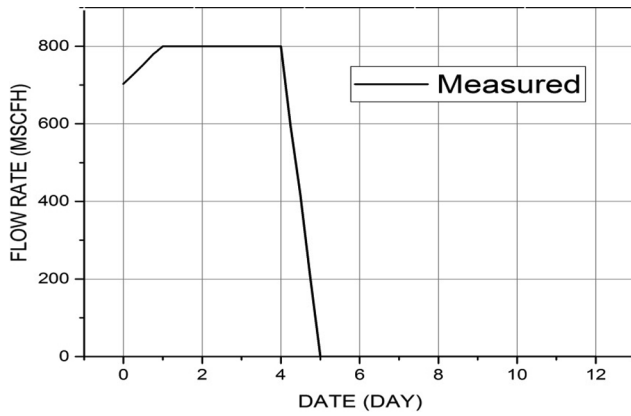


Fig. 2a. The measured gas flowrate during withdrawal test performed in Melville (Canada) cavern [4], and [2].

and mass exchange at the surface of storage facilities. The inner salt cavern is considered as a mass-variation system for thermodynamic analysis.

(1) Gas Withdrawal

In process of gas-withdrawal, gas depressurization leads to gas

Table 1

Parameters used in the calculation for the validation case for withdrawal test performed in Melville (Canada) cavern.

Parameters	Value, Units
Cavern volume (V)	46153 m ³
Cavern surface area (A)	92306 m ²
Wellbore diameter (d ₀)	0.948 m
Gas density (ρ)	0.7143 kg/Sm ³
Constant-pressure specific heat of gas (c _p)	2347 J/(kg·K)
Constant-volume specific heat of gas (c _v)	1848 J/(kg·K)
Initial gas temperature in cavern (T ₀)	305.4 K
Initial gas quality in cavern (M ₁)	3939652 kg
Natural gas constant (R)	519.625 J/(kg·K)
Heat transfer coefficient (h ₀)	30 W/(m ² ·K)
Thermal coefficient K	3 × 10 ⁻⁶
Slope of withdrawal rate ke	9.02 × 10 ⁻⁶
Initial withdrawal rate, ce	5.5 m ³ /s

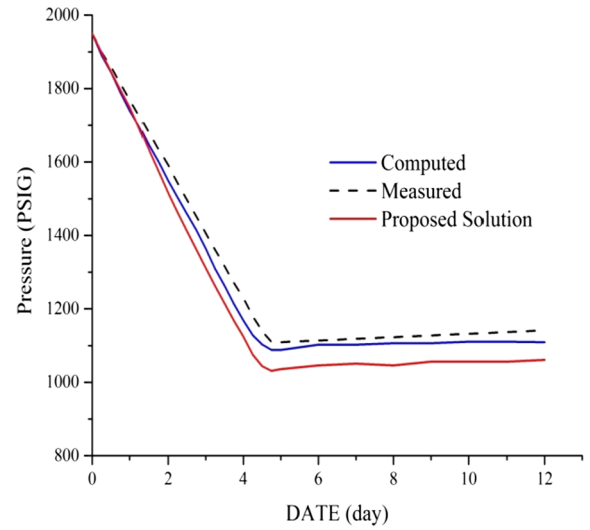


Fig. 2b. Gas pressure evolutions during withdrawal test performed in Melville (Canada) cavern.

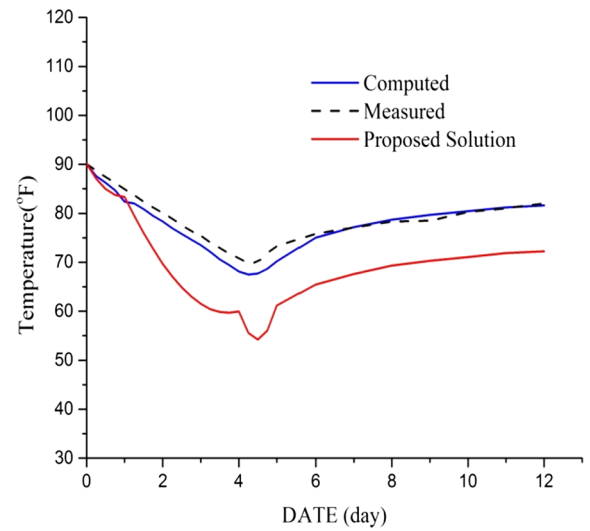


Fig. 2c. Gas temperature evolutions during withdrawal test performed in Melville (Canada) cavern.

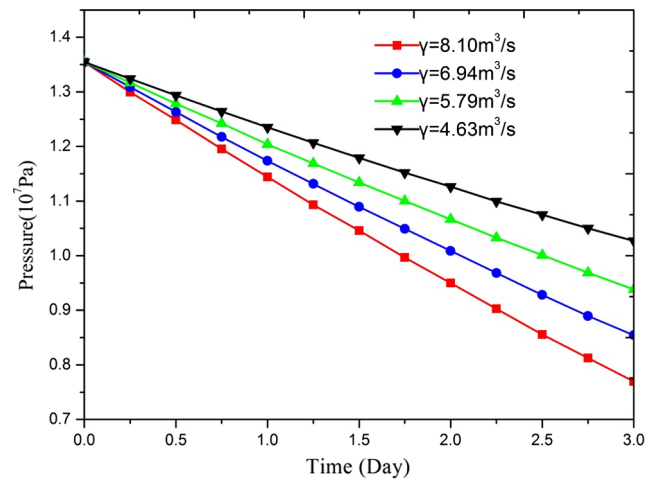


Fig. 3a. Gas pressure variation at different constant withdrawal flowrates.

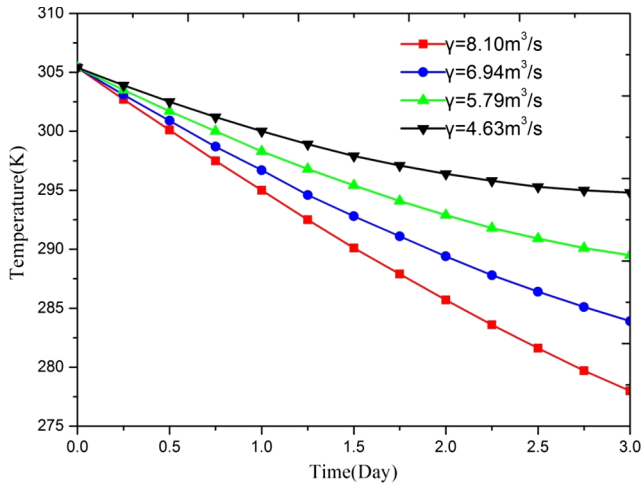


Fig. 3b. Gas temperature variation at different constant withdrawal flowrates.

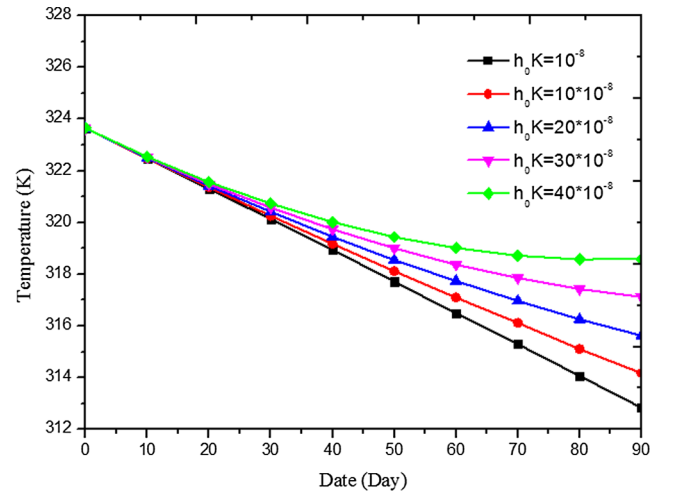


Fig. 5a. Gas temperature variation for different thermal coefficients.

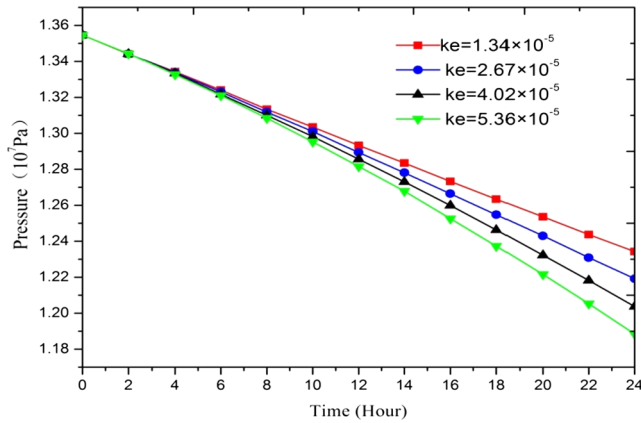


Fig. 4a. Gas pressure variation at different inconstant withdrawal flowrates.

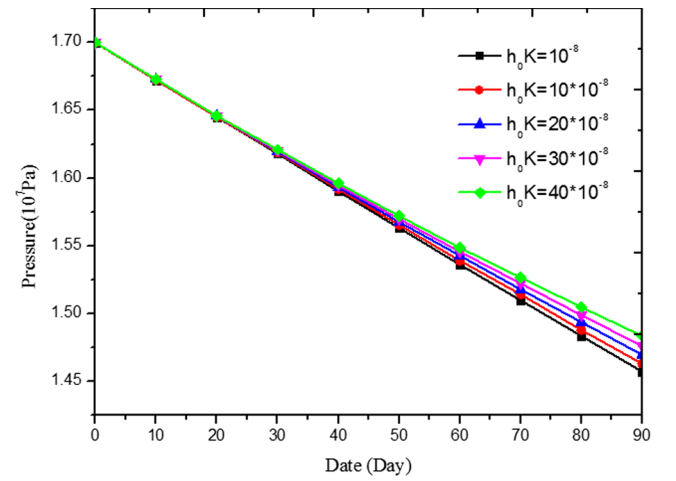


Fig. 5b. Gas pressure variation for different thermal coefficients.

cooling gradually in the cavern. With the gas withdrawing, the induced heat from cavern wall to surrounding rock is increasing, and growing fast at the late stage of gas-withdrawal. The solution can be obtained by procedure in Appendix A. The solution of temperature variation is expressed as following:

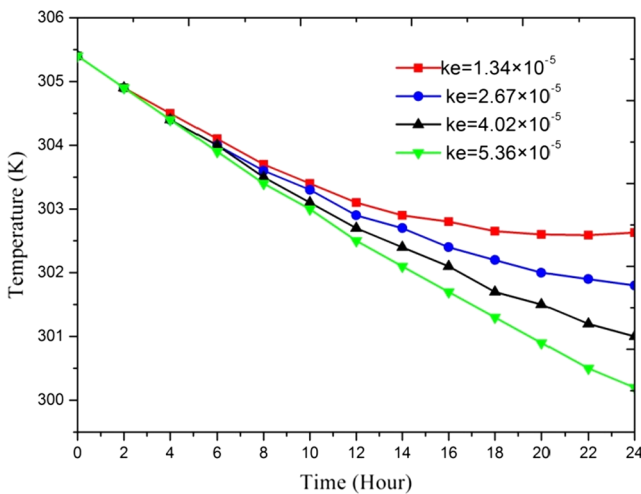


Fig. 4b. Gas temperature variation at different inconstant withdrawal flowrates.

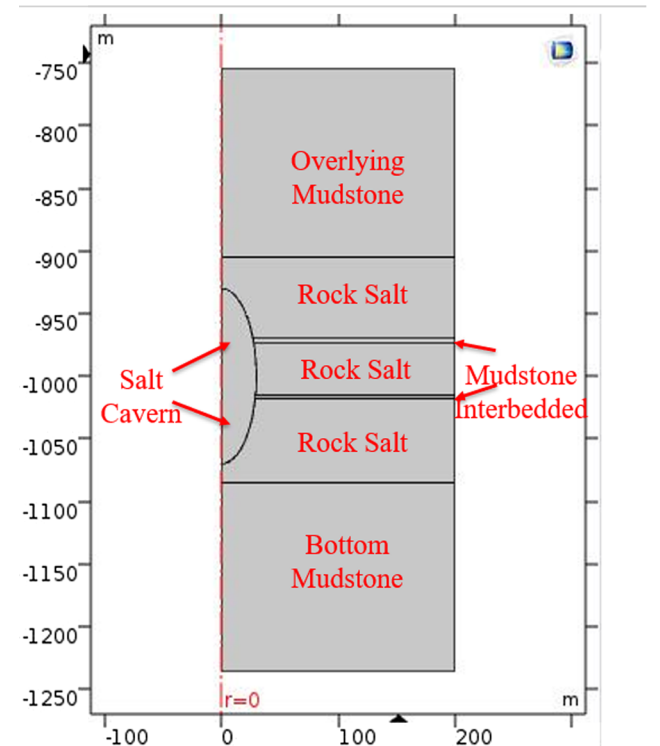


Fig. 6. Scheme for developed numerical model geometry.

Table 2
Rock properties in numerical models.

Parameters	Young's Modulus (GPa)	Poisson's Ratio	Density (kg/m ³)	Thermal conductivity coefficient (W/mK)	Heat capacity (J/kg·K)	Thermal expansion coefficient (K ⁻¹)
Overlying Mudstone	10	0.27	2460	2.8	850	0.0001
Bottom Mudstone	10	0.27	2460	2.8	850	0.0001
Mudstone Interbedded	4	0.3	2350	2.8	850	0.0001
Salt Rock	6	0.3	2160	6.5	837	0.0005

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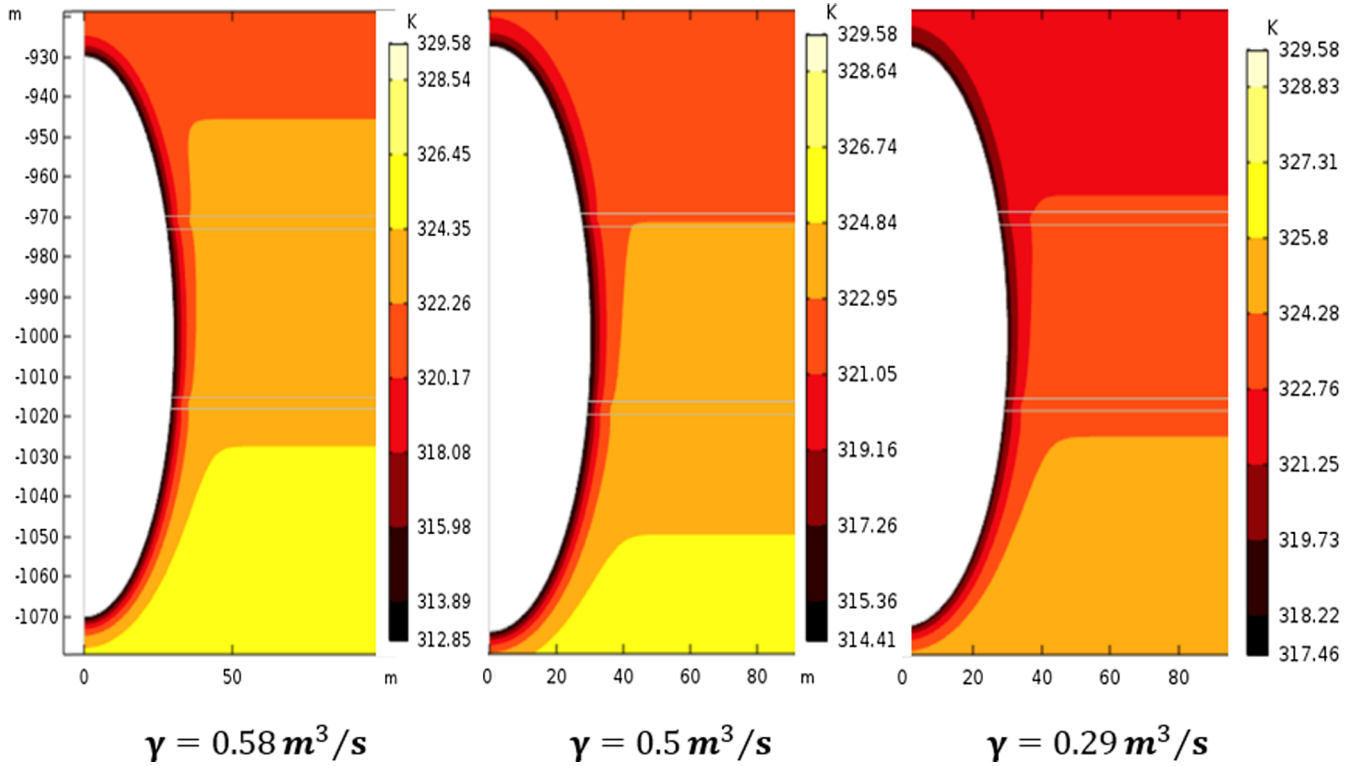


Fig. 7. The rock temperature distribution at 90-day of gas-withdrawal with different gas-withdrawal rates.

$$T = C[(M_1 - \rho\gamma_1\tau)c_v]_{\frac{R}{c_v}} + \frac{10^{-8}A\tau}{\rho\gamma_1 R} - \frac{10^{-8}Ac_v(M_1 - \rho\gamma_1\tau)}{\rho^2\gamma_1^2 R(R - c_v)} - \frac{8\gamma_1^2}{\pi^2 d_0^4 R} \quad (6)$$

Initial boundary:

$$\tau = 0, T = T_{01}$$

Where,

$$C = \left(T_{01} + \frac{8\gamma_1^2}{\pi^2 d_0^4 R} \right) c_v^{-\frac{R}{c_v}} M_1^{-\frac{R}{c_v}} + \frac{10^{-8}Ac_v \frac{c_v - R}{c_v} M_1^{\frac{c_v - R}{c_v}}}{\rho^2\gamma_1^2 R(R - c_v)}$$

The solution of pressure variation is expressed as following:

$$p = -\frac{10^{-8}}{2V}\tau^2 + \frac{8\rho\gamma_1^3}{\pi^2 d_0^4 V}\tau + \frac{CRC_v^{\frac{R}{c_v}}}{V}(M_1 - \rho\gamma_1\tau)^{\frac{c_p}{c_v}} - \frac{10^{-8}Ac_p}{2V\rho^2\gamma_1^2(R - c_v)} \quad (7)$$

Initial boundary:

$$\tau = 0, p = p_{01}$$

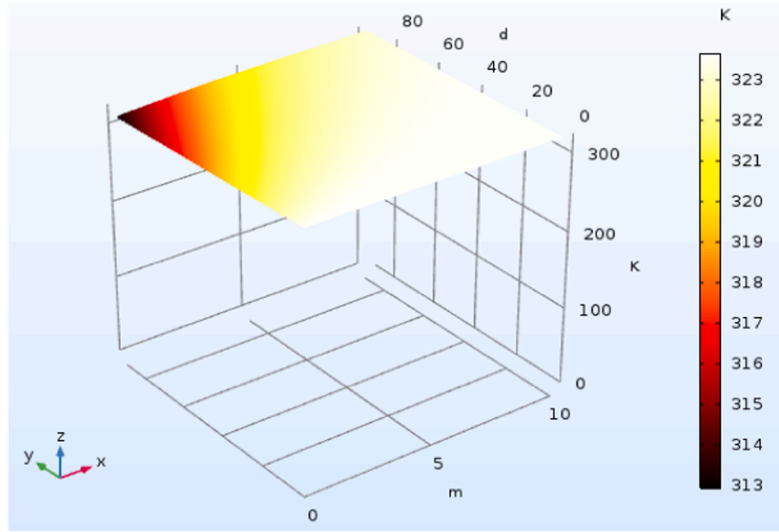
Where,

$$C_2 = p_{01} - \frac{CRC_v^{\frac{R}{c_v}} M_1^{\frac{c_p}{c_v}}}{V} + \frac{10^{-8}Ac_p M_1^2}{2V\rho^2\gamma_1^2(R - c_v)}$$

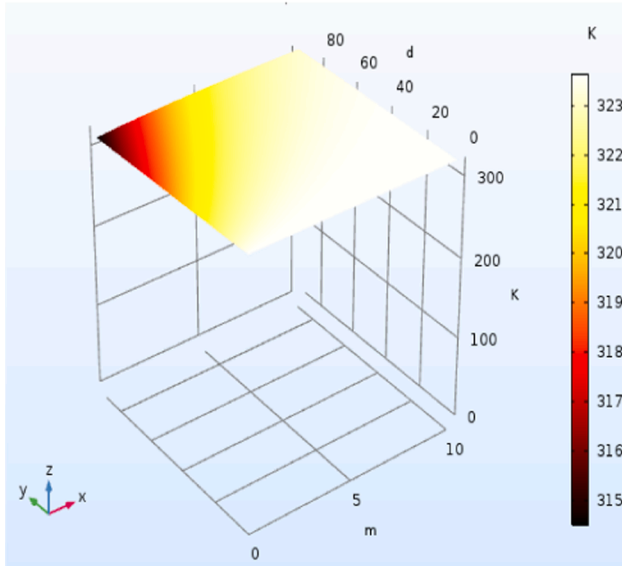
At the condition that air mass flow rate does not remain a constant, when it is linear variation, instead. The solution of temperature variation for un-constant gas-withdrawal rate is expressed as following:

$$\frac{dT}{d\tau} = \frac{h_0 KA\tau - \frac{8(k_e\tau + c_e)^3\rho}{\pi^2 d_0^4} - RT\rho(k_e\tau + c_e)}{[M_1 - \rho\tau(\frac{k_e\tau}{2} + c_e)]c_v} \quad (8)$$

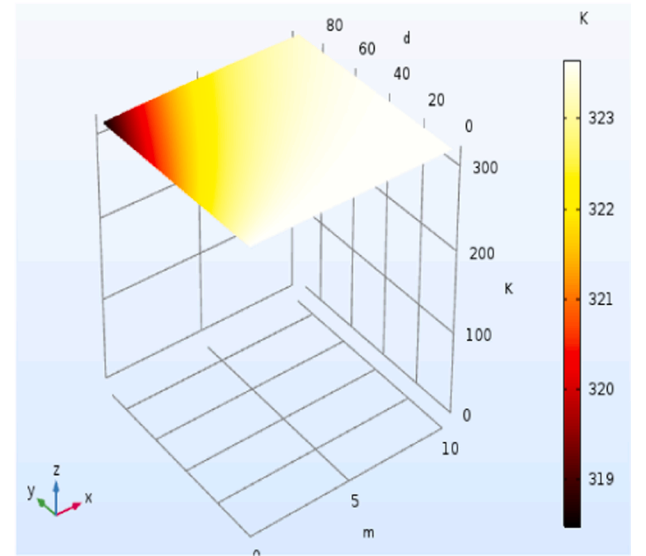
The solution of pressure variation for un-constant gas-withdrawal rate is expressed as following:



$$\gamma = 0.58 \text{ m}^3/\text{s}$$



$$\gamma = 0.5 \text{ m}^3/\text{s}$$



$$\gamma = 0.29 \text{ m}^3/\text{s}$$

Fig. 8. The rock temperature distribution with different gas-withdrawal rate.

$$\frac{dP}{d\tau} = \frac{R}{c_v V} \left[h_0 K A \tau - \left(c_p T + \frac{8(k_e \tau + c_e)^2}{\pi^2 d_0^4} \right) \rho (k_e \tau + c_e) \right] \quad (9)$$

$$T = \frac{1}{M_2 + \rho \gamma_2 \tau} \left[\frac{\rho \gamma_2}{c_v} \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) \tau - \frac{10^{-8} A}{2c_v} \tau^2 + C \right] \quad (10)$$

(2) Gas Injection

With the gas injecting, the gas temperature in cavern, and the cavern wall temperature increase gradually, the induced heat from cavern wall to surrounding rock is growing. There is gradual accretion in the temperature difference between injecting gas and cavern wall, varied with time. The solution can be obtained by procedure in Appendix B. The solution of temperature variation is expressed as following:

Initial boundary:

$$\tau = 0, \quad T = T_{02}$$

Where,

$$C = M_2 T_{02}$$

The solution of pressure variation is expressed as following:

计算结果对比验证

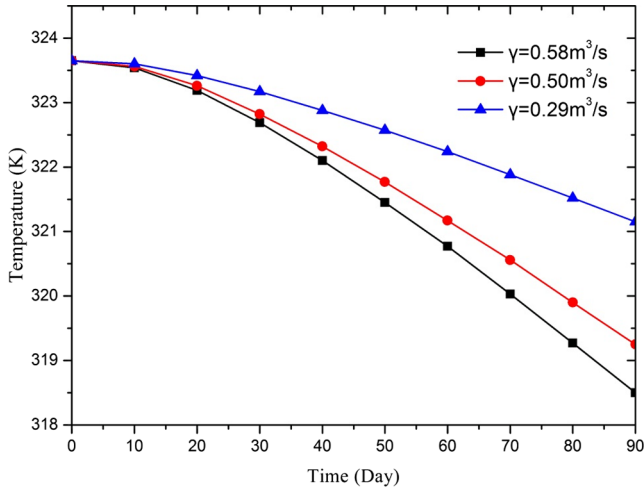


Fig. 9. Temperature at rock salt 3m far away from the carven wall center in process of gas-withdrawal.

$$p = -\frac{10^{-8}AR}{2Vc_v}\tau^2 + \frac{\rho\gamma_2 R}{Vc_v}\left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4}\right)\tau + C_2 \quad (11)$$

Initial boundary:

$$\tau = 0, \quad p = p_{02}$$

Where,

$$C_2 = p_{02}$$

2.2. Model validation

For the validation of the proposed solution to the analytical modeling of gas injection-and-withdrawal process, the cavern gas pressure and temperature variations calculated from the proposed solution were compared with the measured data, and computed results by LMS in Ref. [2]. The gas withdrawal flowrate is demonstrated in Fig. 2(a). The parameters used in the calculation to validate the proposed solution are listed in Table 1. The cavern volume is $V = 46,153 \text{ m}^3$, as the cavern shape is unknown, surface/volume is assigned as a factor of 2, to reach a good fit between measured and computed values [2]. Figs. 2b and 2c illustrate the gas pressure, and temperature calculated from proposed solution, respectively. The pressure of proposed solution is in excellent agreement with measured and computed results in withdrawal test performed in Melville Cavern in Ref. [2]. The temperature of proposed solution shows slight deviation, but still agree with the results from the test. The proposed solution is justifiable theoretically for both constant and time-varying gas-withdrawal rate.

2.3. Parametric analysis

2.3.1. Constant withdrawal rate

Figs. 3a and 3b shows the effect of the air mass flow rate for steady cycle of the salt cavern operation. As observed, with flowrate rising; both the pressure and temperature reduce more rapidly. The drop differences between different flow rates are enlarging with the time. The faster withdrawal rate, the higher pressure or temperature drop would be. Therefore, it is significant to set an appropriate flow rate to avoid over drop in pressure the in the field operation.

2.3.2. Inconstant withdrawal rate

When the air mass flow rate is not a constant, it is time-varying, i.e.

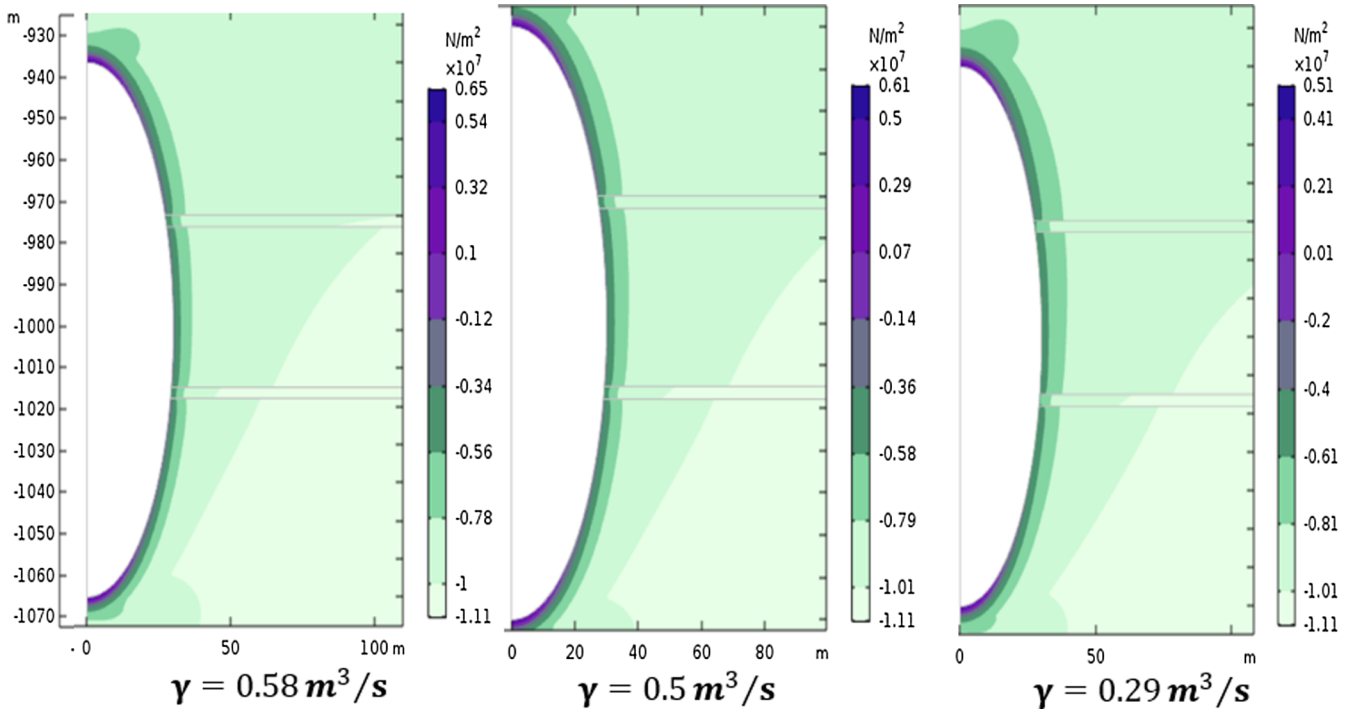


Fig. 10. The cavern wall stress contour at 90-day of gas-withdrawal with different gas-withdrawal rates.

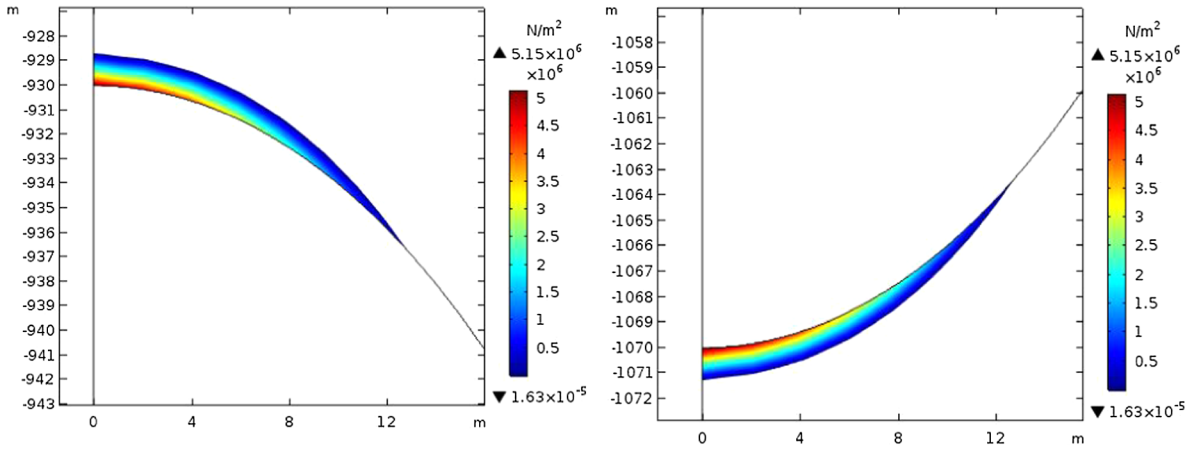


Fig. 11a. Amplification of tensile stress areas in *thermo-mechanical* model at 90-day of gas-withdrawal. $\gamma = 0.29\text{m/s}$

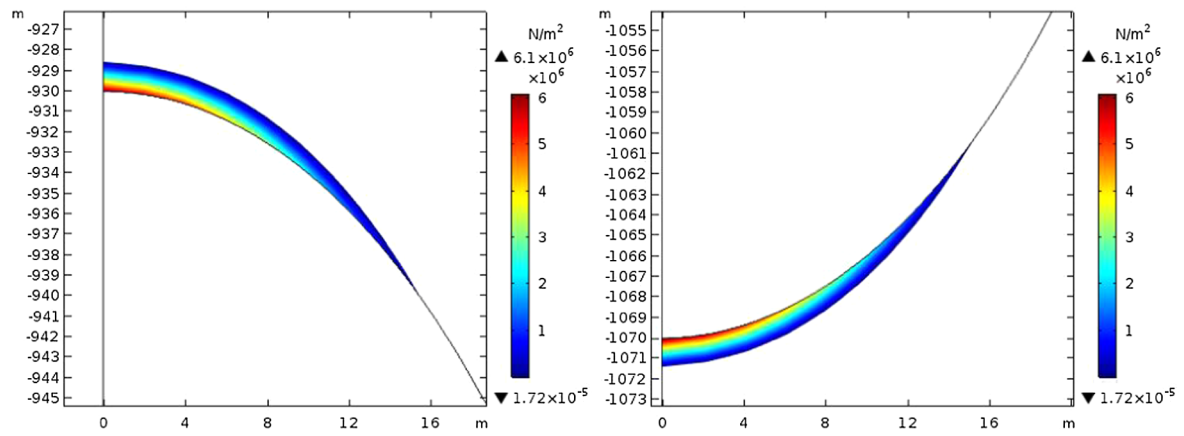


Fig. 11b. Amplification of tensile stress areas in *thermo-mechanical* model at 90-day of gas-withdrawal. $\gamma = 0.58\text{m/s}$

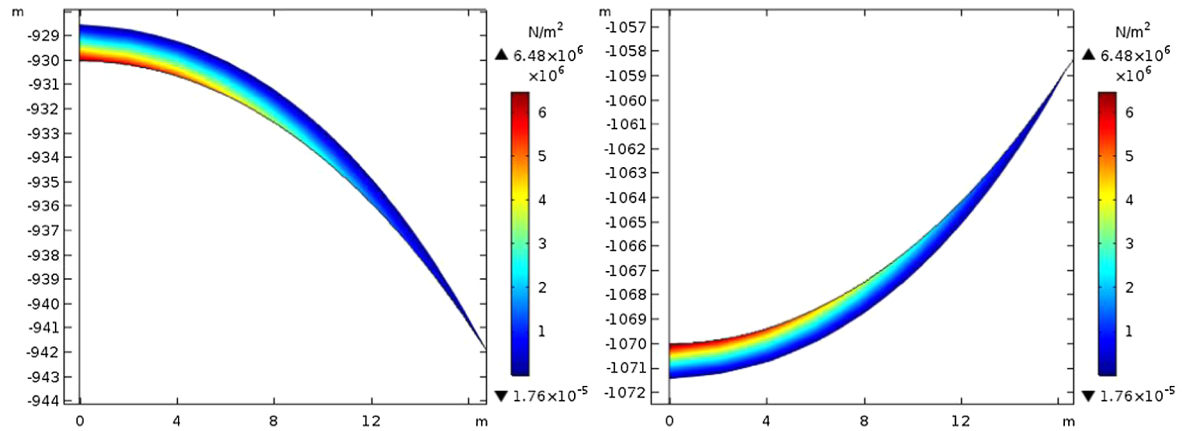


Fig. 11c. Amplification of tensile stress areas in *thermo-mechanical* model at 90-day of gas-withdrawal. $\gamma = 0.87\text{m/s}$

linear variation, instead. Figs. 4a and 4b shows the effect of the air mass flow rate at different slope of rate. As it illustrated, with increasing slope; both the pressure and temperature decrease. The drop magnitude for pressure and temperature become larger with the raising variation of flow rate. In summary, the influence of variation of gas withdrawal rate is significant for both temperature and pressure. Higher flow rate

leads to rapid drop. Therefore, it is imperative to set an appropriate flow rate to avoid over drop in pressure the in the field operation.

2.3.3. Thermal coefficient

h_0k is thermal coefficient that manifests the ability of heat transfer from air in the cavern to the cavern wall. Usually, the gas-withdrawal

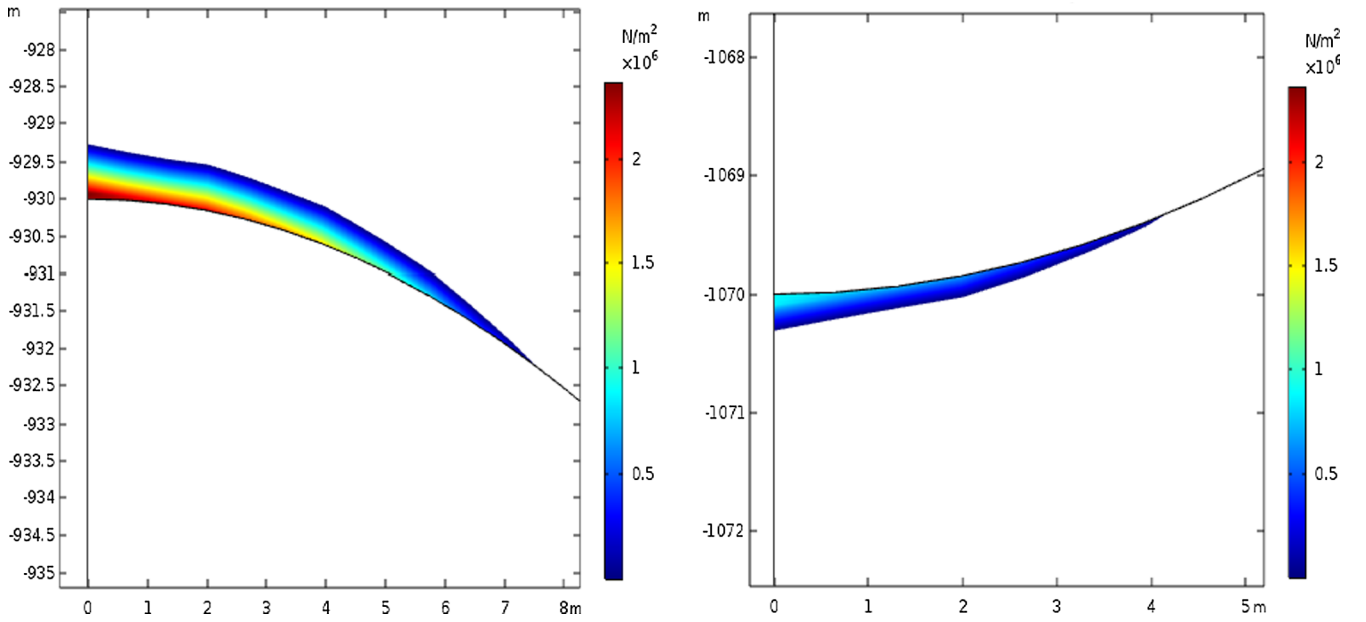


Fig. 12a. Amplification of tensile stress areas in *mechanical model* at 90-day of gas-

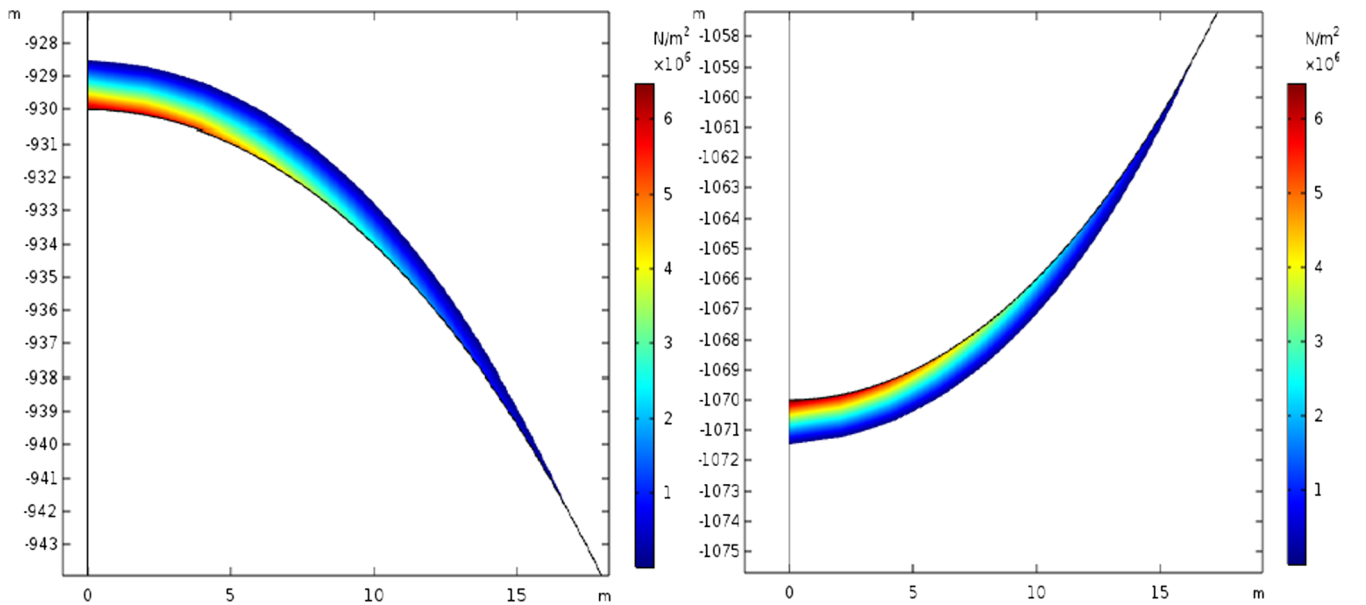


Fig. 12b. Amplification of tensile stress areas in *thermo-mechanical model* at 90-day of gas-withdrawal.

period in winter lasts at least three months, assume the time period of gas-withdrawal at constant rate $\gamma = 0.58 \text{ m}^3/\text{s}$ is 90-day, Figs. 5a and 5b illustrates the variation of temperature and pressure at increasing different values of h_0k . The larger h_0k , the more heat the air could receive from the cavern wall when the air flows out of the cavern, which leads to the slower temperature and pressure drop during process of gas-withdrawal. Therefore, the effect of thermal coefficient can not be neglected in analysis of thermal loading in salt cavern.

3. Thermal response evaluation at salt cavern wall

3.1. Numerical model description

An underground salt cavern model, including flowing air and salt rock formation with interbedded mudstone layers, is built to investigate the thermodynamic response on cavern wall formation during gas injection-and-withdrawal process. In this study, we investigate the salt cavern in Jintan, Jiangsu Province in China. The model of salt cavern is

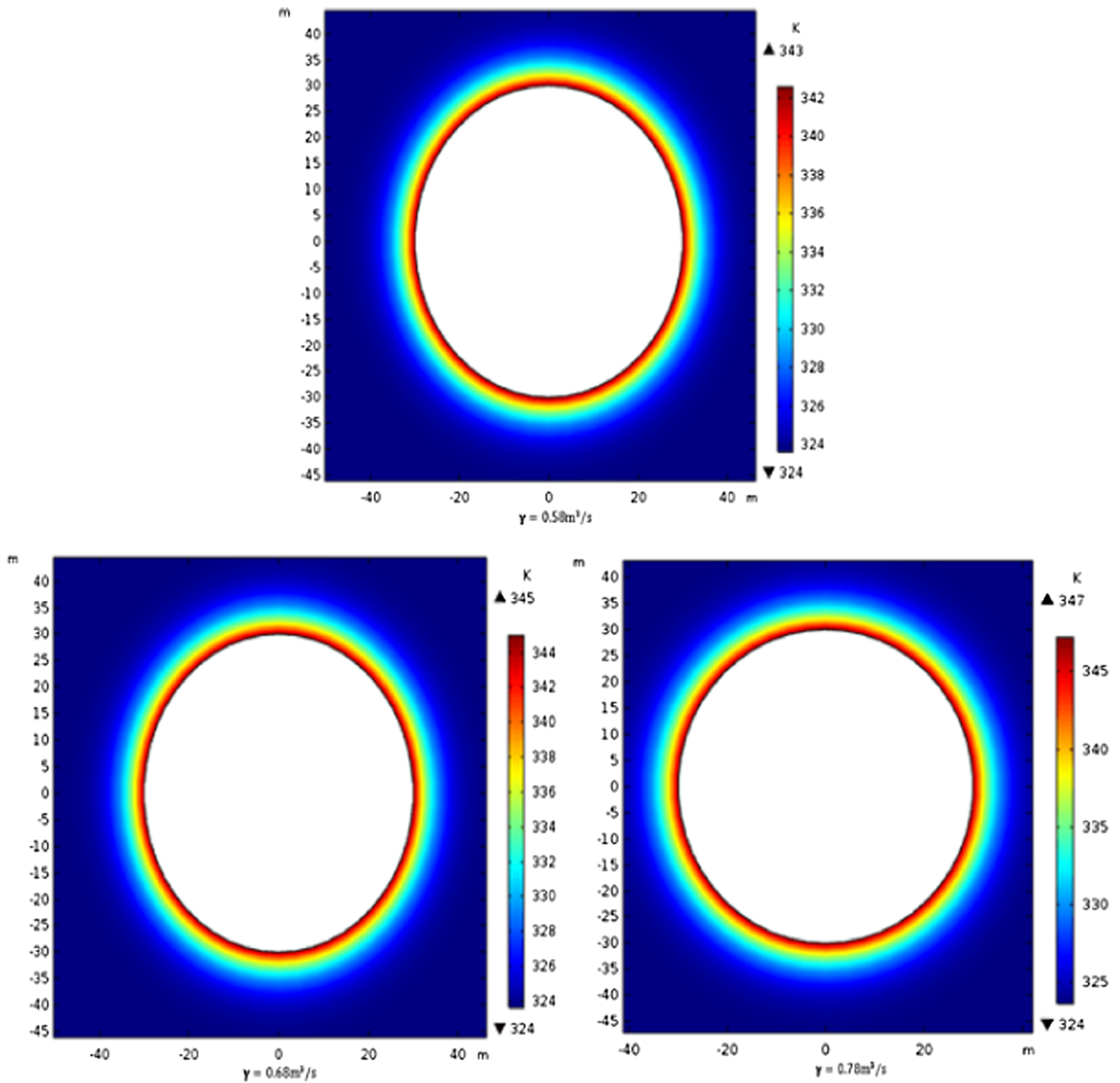


Fig. 13. The cavern wall cross-section temperature distribution with different gas-withdrawal rate.

simplified as spheroid, and symmetric, treating the salt cavern as a simplified cylinder 70 m in elliptical long half axis and 30 m in elliptical short half axis, yielded a total volume of 263,760 m³, the geometric description is shown in Fig. 6. The rock properties are listed in Table 2. The In this study, the free triangle meshes are employed, there are 24,833 domain elements and 868 boundary elements to adapt in the complex configuration. There would be larger stress and temperature gradient at the interface between the air and the surrounding rock, therefore, the boundaries are divided into 8 layers, on the purposes for

grid refinement.

The derived analytical solution for temperature and pressure variation is applied as the boundary conditions for the salt cavern during gas injection-and-withdrawal process. In the numerical model, the salt cavern wall surface allows the heat exchange between air and cavern wall. The formation temperature increases linearly with the depth. The geothermal gradient in this study (e.g. Jintan salt cavern in China) is 2.55 °C/100 m. Initial gas-withdrawal temperature is 323.65 K.

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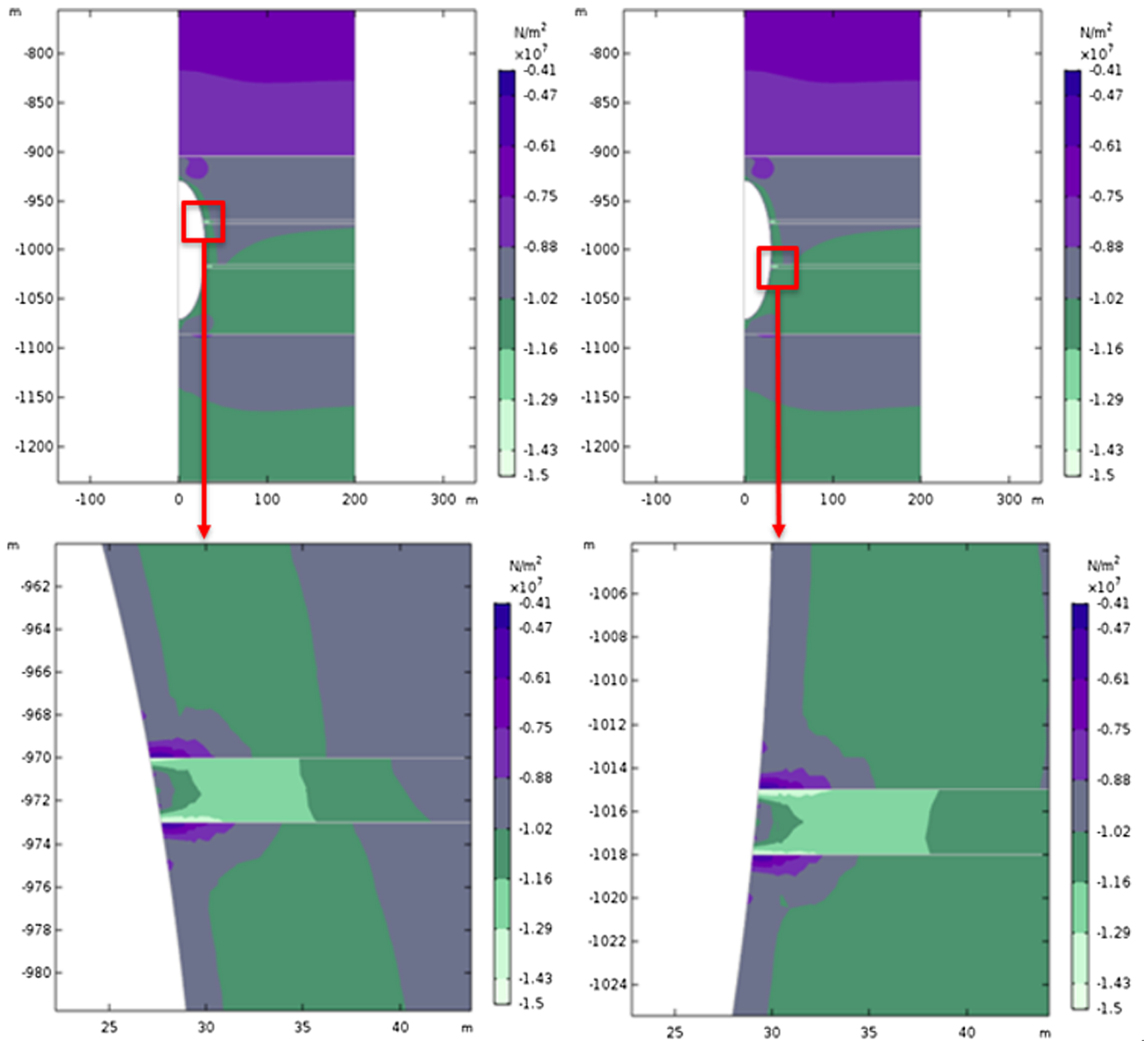


Fig. 14. Maximum principal stress distribution in *thermo-mechanical model* at 90-day of gas-injection at gas-injection rate ($\gamma = 0.78 \text{ m}^3/\text{s}$).

3.2. Thermal response during gas-withdrawal process

During gas emptying periods, the temperature of surrounding rock decrease, which is concurrent with the effective stress increase in the vicinity of salt cavern. In Fig. 7, the temperature distributions at 90-day of gas-withdrawal are compared between different gas-withdrawal rates. It is found that gas-withdrawal rate can have a profound effect on surrounding rock temperature. With the decreasing withdrawal rate, the temperature at cavern wall drops less, heat exchange between the gas and surrounding salt rock mass is slowing down. Fig. 8 shows the

temperature variations at 0to10m distance from the salt cavern wall center over the time period of 90 days with differing gas-withdrawal rates. The temperature fluctuations extend from 0 to 5 m, approximately. Fig. 9 shows that the temperature variation 3m far away from the cavern wall center. The fast gas-withdrawal rates lead to more decline in the temperature of surrounding rock.

Fig. 10 illustrates the stress contour in at 90-day of gas-withdrawal with different gas-withdrawal rates. The tensile stress is positive in magnitude. Figs. 11a–11c illustrates the areas of tensile stress mainly occur on the top and bottom of the cavern. The tensile stress may form

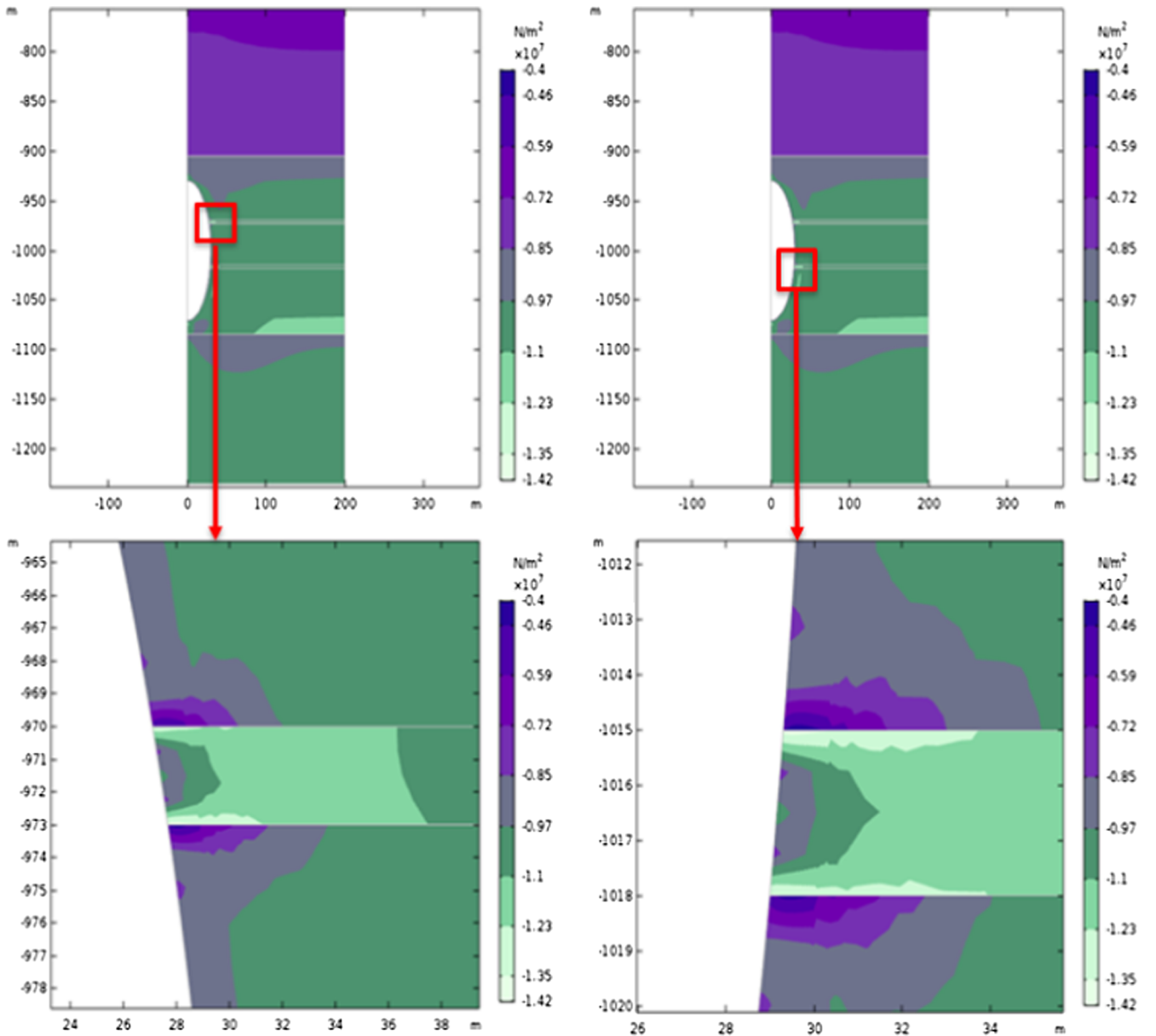


Fig. 15. Maximum principal stress distribution in *thermo-mechanical model* at 90-day of gas-injection at gas-injection rate ($\gamma = 0.68 \text{ m}^3/\text{s}$).

micro-cracks or fractures in formation, resulting in collapse on the roof.

At the gas-withdrawal rate $\gamma = 0.58 \text{ m}^3/\text{s}$, we compare the tensile stress distribution at surrounding rock mass of salt cavern obtained from the mechanical model and from the thermo-mechanical model, shown in Figs. 12a and 12b. There is distinctive difference in the amplification of tensile stress areas and the stress magnitude between simulation results obtained from mechanical model and thermo-mechanical model. The results indicate that in thermo-mechanical model, there are much larger area of tensile stress on top and bottom of salt cavern, and the thermodynamic effects increase the total tensile stress. Therefore, the influence of temperature on surrounding rock mass cannot be neglected in analysis of salt cavern stability.

3.3. Thermal response during gas-injection process

As shown in Fig. 13, the temperature at cavern wall increase with the higher gas-injection rate.

Figs. 14–17 illustrate the maximum principal stress distributions, label and enlarge the areas of the upper and lower intersections between mudstone layer and rock salt layers. There are no areas of tensile stress in process of gas filling into salt cavern. Fig. 14, Fig. 15, and Fig. 16 show that, with the elevation of gas-injection rate, the stress magnitude increase. Particularly, the areas of lower stress concentrate on interaction between rock salt layer and mudstone layer. In the process of gas-injection, the intervention of interbedded layers of rock

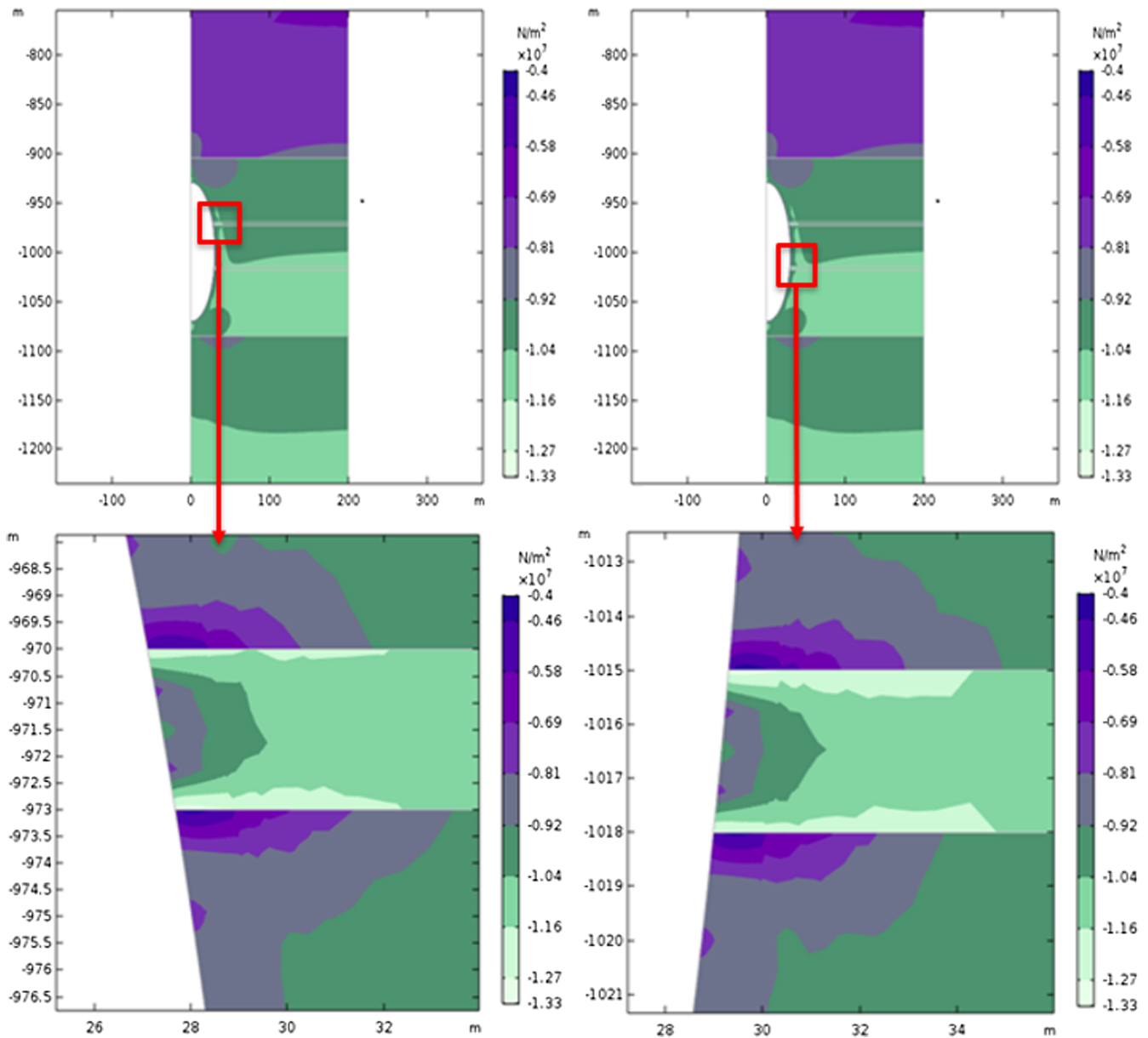


Fig. 16. Maximum principal stress distribution in *thermo-mechanical* model at 90-day of gas-injection at gas-injection rate ($\gamma = 0.58 \text{ m}^3/\text{s}$).

salts formation may cause the disturbance of stress distribution.

We compare the maximum principal stress distribution at surrounding rock mass of salt cavern obtained from the mechanical model and thermo-mechanical model for the same gas injection rate $\gamma = 0.58 \text{ m}^3/\text{s}$, shown in Fig. 17, and Fig. 16, respectively. There is obvious difference in stress field obtained from between thermo-mechanical model and mechanical model. The thermodynamic effects decrease the stress magnitude.

Fig. 18 shows that the temperature variation with the time 3m far away from the cavern wall center. The fast injection results in the elevation of surrounding rock temperature.

4. Results and discussion

The derived solution can be used in the thermodynamic analysis of salt cavern operation, as the boundary conditions for calculation of thermal and stress field. The variation of different withdrawal and injection rates causes significant temperature and stress distribution changes in rock salt mass. In process of gas-withdrawal, tensile stresses calculated taking thermal effect into account in thermo-dynamic model are larger than those calculated by mechanical model. The thermal induced stresses in rock salt are in the form of tensile stresses, which mainly concentrate in the region of top and bottom of the cavern. These

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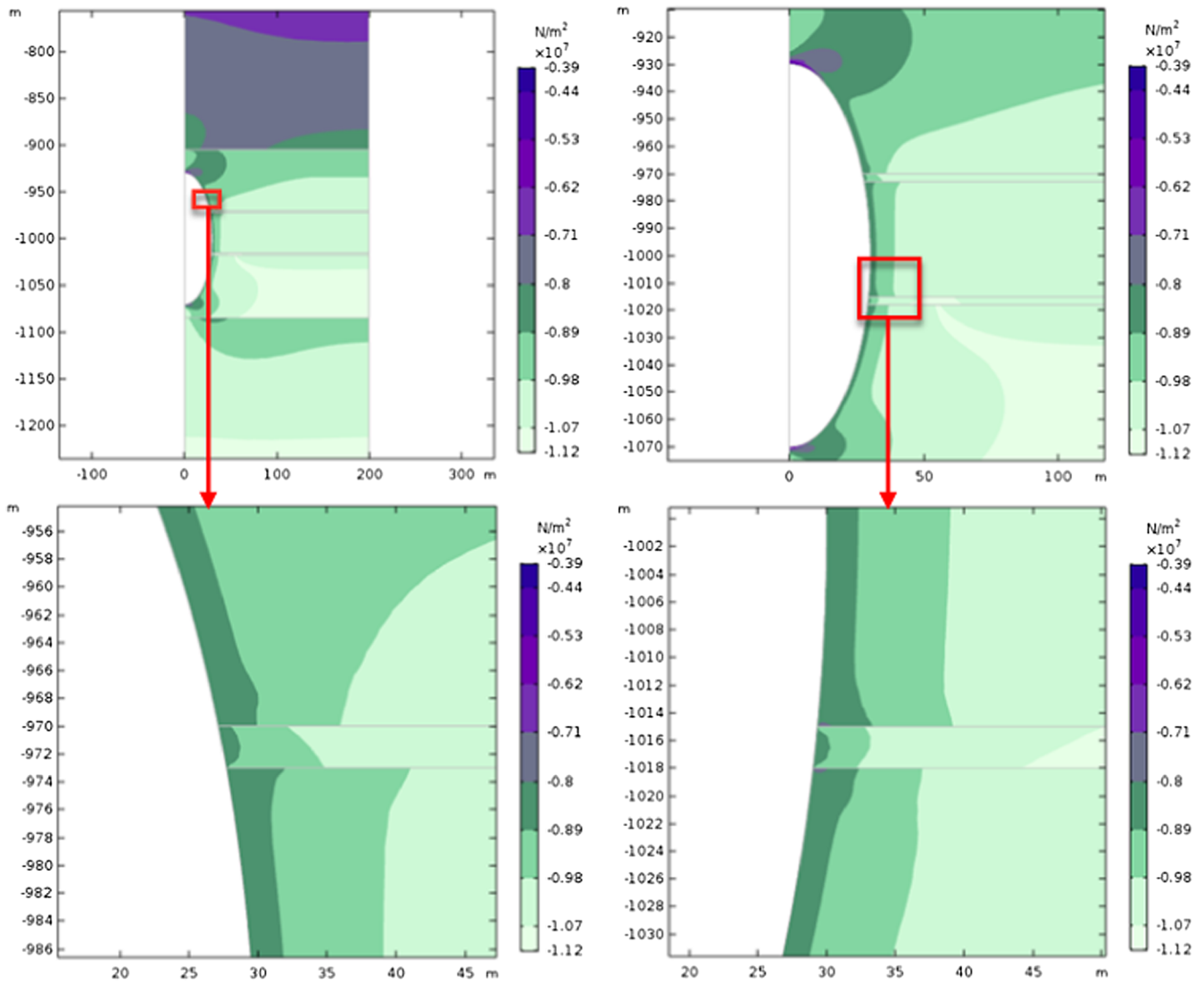


Fig. 17. Maximum principal stress distribution in *mechanical model* at 90-day of gas-injection at gas-injection rate ($\gamma = 0.58 \text{ m}^3/\text{s}$).

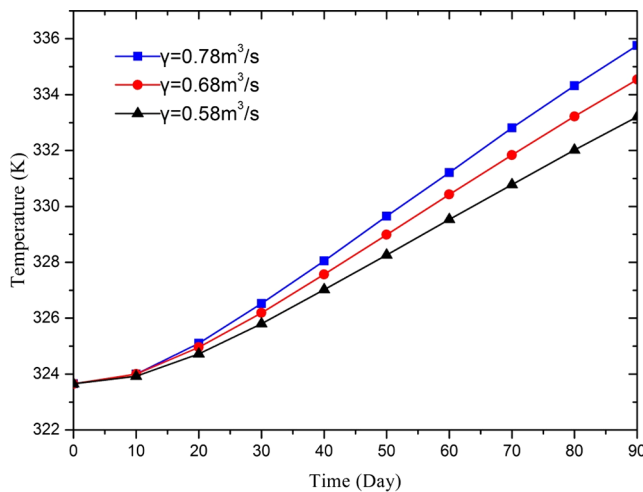


Fig. 18. Temperature at rock salt 3m far away from the cavern wall center in process of gas-injection.

tensile stresses would contribute to the generation of thermal cracks during gas-withdrawal. Fast gas withdrawal leads to the temperature drop and larger tensile areas on the top and bottom of the cavern;

inversely, fast gas-filling results in temperature increase. Although there are no tensile area occurs in gas-injection, there is obvious stress disturbance at the interface between salt formation and the interbedded mudstone layers. The intervention of interlayers has more impacts in process of gas-injection, rather than in the process of gas-withdrawal. Therefore, the impact of thermodynamics in cavern operation scenarios should be considered to determine the operation parameters precisely, avoid the roof collapses due to effect of gravity in the cavern induced by fast withdrawal or injection.

5. Conclusion

Due to seasonal storage, the cycling scenarios of underground salt cavern operation leads to changes both in temperature and pressure of gas inside the cavern, resulting in thermal effects on the surrounding rock formation in vicinity of the cavern.

The proposed mathematical model is validated by withdrawal test performed in Melville Cavern, and the model is capable of calculating thermal stress at cavern wall induced during gas injection-and-withdrawal processes.

The results indicate that there are differences in thermo-dynamic response between gas filling and emptying. During gas withdrawal period, tensile stresses increase, and there are distinctive tensile stress concentration areas on the roof and floor of the cavern; while in the gas-

injection period, there are no tensile stress area, however, it inclines to occur stress disturbance at intervention of interlayers.

Parameters analysis shows that both the thermal coefficients and variations of injection-and-withdrawal rates have impact on the thermodynamic responses of surrounding rock mass. Fast injection or withdrawal results in severe changes in both temperature and pressure of air in cavern, induces surrounding rock temperature changes, and thermal stress.

Therefore, the thermodynamic effect should be included into

evaluation of cavern stability at different operation conditions. This study provides reference for optimal determination of operation parameters, and design criteria of salt cavern as well.

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Appendix A. Solutions of temptation and pressure variation for gas-withdrawal process

At the beginning of gas-withdrawal, in salt cavern, the gas initial temperature is T_{01} , initial pressure is p_{01} , the initial mass of gas is M_1 . when the gas-withdrawal rate is γ_1 , in the time period of τ , the mass of gas-withdrawal is $\rho\gamma_1\tau$, the residual mass of gas in salt cavern is $M_1 - \rho\gamma_1\tau$, as following:

$$\frac{\delta m}{d\tau} = -\rho\gamma_1, m = M_1 - \rho\gamma_1\tau \quad (\text{A-1})$$

in salt cavern, heat exchange between gas in salt cavern and salt rock mass surrounding per unit time is expressed as:

$$\frac{\delta Q}{d\tau} = h_0 A k \tau \quad (\text{A-2})$$

where, k is coefficient of temperature difference between air and cavern wall, h_0 is the heat transfer coefficient, $h_0 k = 10^{-8}$ [16]

$$\frac{\delta Q}{d\tau} = 10^{-8} A \tau \quad (\text{A-3})$$

Enthalpy for ideal gas:

$$H = mh = mc_p T \quad (\text{A-4})$$

During gas-withdrawal, the relation between air temperature inside the cavern and time is:

$$\frac{dT}{d\tau} = \left[\frac{\delta Q}{d\tau} + \left(h + \frac{v_r^2}{2} \right) \frac{\delta m}{d\tau} - T c_v \frac{\delta m}{d\tau} \right] / (mc_v) \quad (\text{A-5})$$

$$v_r = \frac{M}{F\rho} = \frac{4M}{\pi d_0^2 \rho} = \frac{4\gamma}{\pi d_0^2} \quad (\text{A-6})$$

where, v_r is the air flow rate in vertical wellbore, m/s; M is the air mass flow rate in vertical wellbore, kg/s; F is the circulation cross-section area, m²; d_0 is the inner diameter of vertical wellbore, m

Plug (A-1), (A-4), (A-6), into (A-5) gives:

$$\frac{dT}{d\tau} = \frac{10^{-8} A \tau + \left(c_p T + \frac{8\gamma_1^2}{\pi^2 d_0^4} \right) (-\rho\gamma_1) - c_v T (-\rho\gamma_1)}{(M_1 - \rho\gamma_1\tau) c_v} \quad (\text{A-7})$$

For ideal gas, the relation between constant volume specific heat c_v and constant volume specific heat c_p is:

$$c_p - c_v = R \quad (\text{A-8})$$

it follows from Eqs. (A-7) and (A-8) that:

$$\frac{dT}{d\tau} + \frac{R\rho\gamma_1}{(M_1 - \rho\gamma_1\tau)c_v} T = \frac{10^{-8} A \tau - \frac{8\rho\gamma_1^3}{\pi^2 d_0^4}}{(M_1 - \rho\gamma_1\tau)c_v} \quad (\text{A-9})$$

Eq. (A-9) is nonhomogeneous linear equation, to solve that equation, the corresponding homogeneous linear equation of Eq. (A-9) needs to be solved first, i.e.:

$$\frac{dT}{d\tau} + \frac{R\rho\gamma_1}{(M_1 - \rho\gamma_1\tau)c_v} T = 0 \quad (\text{A-10})$$

Eq. (A-10) separate variables, it gives:

$$\frac{dT}{d\tau} = -\frac{R\rho\gamma_1}{(M_1 - \rho\gamma_1\tau)c_v} T \quad (\text{A-11})$$

Integrate both sides, it arrives:

$$\ln T = \frac{R}{c_v} \ln [(M_1 - \rho\gamma_1\tau)c_v] + C_1 \quad (\text{A-12})$$

which is:

$$T = C [(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R}{c_v}} \quad (\text{A-13})$$

Eq. (A-13) is the general solution of corresponding homogeneous linear Eq. (A-10) using Constance Variation method to solve the general solution of non-homogeneous linear Eq. (A-9), replace C into unknown function $u(\tau)$ of τ in Eq. (A-13), it gives:

$$T = u(\tau)[(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R}{c_v}} \quad (\text{A-14})$$

then:

$$\frac{dT}{d\tau} = u'(\tau)[(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R}{c_v}} - (\rho\gamma_1 R)[(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R-c_v}{c_v}} u(\tau) \quad (\text{A-15})$$

Plug Eq. (A-14), and Eq. (A-15) into Eq. (A-9), it gives:

$$u'(\tau)[(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R}{c_v}} = \frac{10^{-8}A\tau - \frac{8\rho\gamma_1^3}{\pi^2 d_0^4}}{(M_1 - \rho\gamma_1\tau)c_v} \quad (\text{A-16})$$

i.e.:

$$u'(\tau) = (10^{-8}A\tau - \frac{8\rho\gamma_1^3}{\pi^2 d_0^4})[(M_1 - \rho\gamma_1\tau)c_v]^{-\frac{c_p}{c_v}} \quad (\text{A-17})$$

Integrate both sides of the above equation, it arrives:

$$u(\tau) = \int (10^{-8}A\tau - \frac{8\rho\gamma_1^3}{\pi^2 d_0^4})[(M_1 - \rho\gamma_1\tau)c_v]^{-\frac{c_p}{c_v}} d\tau + C \quad (\text{A-18})$$

use partial integration to above formula, we obtain the following:

$$u(\tau) = 10^{-8}Ac_v^{-\frac{c_p}{c_v}} \left[\frac{c_v\tau(M_1 - \rho\gamma_1\tau)^{-\frac{R}{c_v}}}{\rho\gamma_1 R} - \frac{c_v^2(M_1 - \rho\gamma_1\tau)^{-\frac{R-c_v}{c_v}}}{\rho^2\gamma_1^2 R(R - c_v)} - \frac{8\gamma_1^2 c_v}{\pi^2 d_0^4 R} c_v^{-\frac{c_p}{c_v}} (M_1 - \rho\gamma_1\tau)^{-\frac{R}{c_v}} + C \right] \quad (\text{A-19})$$

Plug Eq. (A-19) into Eq. (A-14), it gives:

$$T = C[(M_1 - \rho\gamma_1\tau)c_v]^{\frac{R}{c_v}} + \frac{10^{-8}A\tau}{\rho\gamma_1 R} - \frac{10^{-8}Ac_v(M_1 - \rho\gamma_1\tau)}{\rho^2\gamma_1^2 R(R - c_v)} - \frac{8\gamma_1^2}{\pi^2 d_0^4 R} \quad (\text{A-20})$$

Initial boundary conditions:

$$\tau = 0, T = T_{01} \quad (\text{A-21})$$

Plug Eq. (A-21) into Eq. (A-20), it gives C:

$$C = \left(T_{01} + \frac{8\gamma_1^2}{\pi^2 d_0^4 R} \right) c_v^{-\frac{R}{c_v}} M_1^{-\frac{R}{c_v}} + \frac{10^{-8}Ac_v^{\frac{c_v-R}{c_v}} M_1^{\frac{c_v-R}{c_v}}}{\rho^2\gamma_1^2 R(R - c_v)} \quad (\text{A-22})$$

Eq. (A-20) is the solution to gas temperature variation changing with time in the process of gas-withdrawal During gas-withdrawal, the relation between air pressure inside the cavern and time is:

$$\frac{dp}{d\tau} = \left[\frac{\delta Q}{d\tau} + \left(h + \frac{v_z^2}{2} \right) \frac{\delta m}{d\tau} \right] / \left(\frac{V}{R} c_v \right) \quad (\text{A-23})$$

Plug Eq. (A-1), Eq. (A-4), Eq. (A-6) and Eq. (A-3) into Eq. (A-23), it gives:

$$\frac{dp}{d\tau} = \frac{R}{Vc_v} [10^{-8}A\tau + (c_p T + \frac{8\gamma_1^2}{\pi^2 d_0^4})(-\rho\gamma_1)] \quad (\text{A-24})$$

Plug Eq. (A-20) into Eq. (A-24) separate variables, and integrate, it gives:

$$p = -\frac{10^{-8}A}{2V}\tau^2 + \frac{8\rho\gamma_1^3}{\pi^2 d_0^4 V}\tau + \frac{CRc_v^{\frac{R}{c_v}}}{V}(M_1 - \rho\gamma_1\tau)^{\frac{c_p}{c_v}} - \frac{10^{-8}Ac_p}{2V\rho^2\gamma_1^2(R - c_v)}(M_1 - \rho\gamma_1\tau)^2 + C_2 \quad (\text{A-25})$$

Initial boundary conditions:

$$\tau = 0, p = p_{01} \quad (\text{A-26})$$

Plug Eq. (A-26) into Eq. (A-25), it gives C_2 :

$$C_2 = p_{01} - \frac{CRc_v^{\frac{R}{c_v}} M_1^{\frac{c_p}{c_v}}}{V} + \frac{10^{-8}Ac_p M_1^2}{2V\rho^2\gamma_1^2(R - c_v)} \quad (\text{A-27})$$

Eq. (A-25) is the solution to gas pressure variation changing with time in the process of gas-withdrawal

Appendix B. Solutions of temptation and pressure variation for gas injection process

At the beginning of gas-injection, in salt cavern, the gas initial temperature is T_{02} , initial pressure is p_{02} , the initial injected-gas temperature is constant T_{in} , mass of gas is M_2 . when the gas-injection rate is γ_2 , in the time period of τ , the mass of gas-withdrawal is $\rho\gamma_2\tau$, the total mass of gas in salt cavern is $M_2 + \rho\gamma_2\tau$, as following:

$$\frac{\delta m}{d\tau} = -\rho\gamma_1, \quad m = M_1 - \rho\gamma_1\tau \quad (\text{B-1})$$

in salt cavern, heat exchange between gas in salt cavern and salt rock mass surrounding per unit time is expressed as:

$$\frac{\delta Q}{d\tau} = -h_0 A k \tau \quad (\text{B-2})$$

where, the negative sign indicates the heat release induced by the injected gas to the surrounding rock mass,

k is coefficient of temperature difference between air and cavern wall, h_0 is the heat transfer coefficient, $h_0 k = 10^{-8}$ [16]

the exchanged heat between air and surrounding rock is:

$$\frac{\delta Q}{d\tau} = -10^{-8} A \tau \quad (\text{B-3})$$

Plug Eq. (A-4), Eq. (A-6), Eq. (B-1) and Eq. (B-3) into Eq. (A-5), it gives:

$$\frac{dT}{d\tau} = \frac{-10^{-8} A \tau + \rho\gamma_2 \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) - c_v T \rho\gamma_2}{(M_2 + \rho\gamma_2 \tau) c_v} \quad (\text{B-4})$$

which is:

$$\frac{dT}{d\tau} + \frac{\rho\gamma_2}{M_2 + \rho\gamma_2 \tau} T = \frac{\rho\gamma_2 \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) - 10^{-8} A \tau}{(M_2 + \rho\gamma_2 \tau) c_v} \quad (\text{B-5})$$

Eq. (B-5) is nonhomogeneous linear equation, to solve that equation, the corresponding homogeneous linear equation of Eq. (B-5) needs to be solved first, i.e.:

$$\frac{dT}{d\tau} + \frac{\rho\gamma_2}{M_2 + \rho\gamma_2 \tau} T = 0 \quad (\text{B-6})$$

Eq. (B-6) separate variables, it gives:

$$\frac{dT}{T} = -\frac{\rho\gamma_2 d\tau}{M_2 + \rho\gamma_2 \tau} \quad (\text{B-7})$$

integrate both sides of the equation, it gives:

$$\ln T = -\ln(M_2 + \rho\gamma_2 \tau) + C_1 \quad (\text{B-8})$$

then:

$$T = \frac{C}{M_2 + \rho\gamma_2 \tau} \quad (\text{B-9})$$

Eq. (B-9) is the general solution of corresponding homogeneous linear Eq. (B-9) using Constance Variation method to solve the general solution of non-homogeneous linear Eq. (B-8), replace C into unknown function $u(\tau)$ of τ in Eq. (B-9), it gives:

$$T = \frac{u(\tau)}{M_2 + \rho\gamma_2 \tau} \quad (\text{B-10})$$

then:

$$\frac{dT}{d\tau} = \frac{u'(\tau)}{M_2 + \rho\gamma_2 \tau} - \frac{\rho\gamma_2}{(M_2 + \rho\gamma_2 \tau)^2} u(\tau) \quad (\text{B-11})$$

Plug Eq. (B-10), and Eq. (B-11) into Eq. (B-5), it gives:

$$u'(\tau) = \frac{\rho\gamma_2}{c_v} \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) - \frac{10^{-8} A \tau}{c_v} \quad (\text{B-12})$$

integrate both sides of the equation, it gives:

$$u(\tau) = \frac{\rho\gamma_2}{c_v} \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) \tau - \frac{10^{-8} A}{2c_v} \tau^2 + C \quad (\text{B-13})$$

Plug Eq. (B-13) into Eq. (B-10), it gives:

$$T = \frac{1}{M_2 + \rho\gamma_2 \tau} \left[\frac{\rho\gamma_2}{c_v} \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) \tau - \frac{10^{-8} A}{2c_v} \tau^2 + C \right] \quad (\text{B-14})$$

Initial boundary conditions:

$$\tau = 0, \quad T = T_{02} \quad (\text{B-15})$$

It follows from Eq. (B-15), and Eq. (B-14), obtain C :

$$C = M_2 T_{02} \quad (\text{B-16})$$

Eq. (B-14) is the solution to gas temperature variation changing with time in the process of gas-injection

Plug Eq. (A-4), Eq. (A-6), Eq. (B-1) and Eq. (B-3) into Eq. (A-23), it gives:

$$\frac{dp}{d\tau} = \frac{R}{Vc_v} \left[-10^{-8} A \tau + \rho \gamma_2 \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) \right] \quad (\text{B-17})$$

Eq. (B-17) separate variables, and integrate:

$$p = -\frac{10^{-8} AR}{2Vc_v} \tau^2 + \frac{\rho \gamma_2 R}{Vc_v} \left(c_p T_{in} + \frac{8\gamma_2^2}{\pi^2 d_0^4} \right) \tau + C_2 \quad (\text{B-18})$$

Initial boundary conditions:

$$\tau = 0, p = p_{02} \quad (\text{B-19})$$

Plug Eq. (B-19) into Eq. (B-18), it gives C_2 :

$$C_2 = p_{02} \quad (\text{B-20})$$

Eq. (B-18) is the solution to gas pressure variation changing with time in the process of gas-injection

Appendix C. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.114380>.

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